Mohammad Montazeri

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Education

Sharif University of Technology **Tehran, Iran**
M.Sc. in Mechanical Engineering — GPA: 16.5/20 (3.6/4.0) *Oct. 2025 – Present*
Thesis: "Computer-Vision-Based Estimation of Fine Hand Motion for Pick-and-Place Tasks in Telerehabilitation"

University of Tehran **Tehran, Iran**
B.Sc. in Mechanical Engineering — GPA: 17.0/20 (3.6/4.0) *Oct. 2020 – Sep. 2025*
Thesis: "Kinematic Modeling and Redundancy Resolution of Bio-Inspired Gough-Stewart Robots via Deep Learning"

Research Interests

• Robotics and Mechatronics • Biomechanics • Rehabilitation • Computer Vision and AI

Publications

Redundancy Resolution of Rehabilitative Multi-Segment Gough-Stewart Platforms *Accepted*
M. Montazeri, P. Namazian, and M. Tale Masouleh. Submitted to the 33rd National and 11th International Iranian Conference on Biomedical Engineering (ICBME 2026).

Kinematic Modeling of Bio-Inspired Hybrid Gough-Stewart Manipulator *Dec. 2025*
M. Montazeri, P. Namazian, M. Tale Masouleh, M. Nayeri, and A. Kalhor. Presented at the 13th RSI International Conference on Robotics and Mechatronics (ICRoM 2025).

Programming with Python: A Self-Study Tutorial Book *Under review*
Author and editor of a 170-page English-language introductory programming book.

Research Experience

Medical Robotics Laboratory **Sharif University of Technology**
Research Assistant — Supervisor: Prof. Saeed Behzadipour *2025 – Present*
– Developing a computer-vision motion-capture pipeline for fine motion detection in telerehabilitation.
– Integrating the pipeline with biomechanical models to simulate physically feasible grasps for neurological patients

Human and Robot Interaction Laboratory (TaarLab) **University of Tehran**
Research Assistant — Supervisor: Prof. Mehdi Tale Masouleh *2023 – 2025*
– Modeled the kinematics of serially connected Gough-Stewart platforms using deep neural networks.
– Developed redundancy-resolution methods for trajectory tracking and obstacle avoidance in biomimetic robots.
– Resulted in a Bachelor's thesis and two conference manuscripts

Teaching Experience

Robotics Tutor: Sigma Educational Group *Jul. 2025 - Jul. 2026*
– Taught electronics, microcontroller programming, and robotics concepts to high school students.
– Mentored student teams through hands-on projects to build autonomous robots using C++ and Arduino.

Teaching Assistant: University of Tehran
– Designed homework assignments and course projects. – Prepared course materials and graded assignments.

- **Design of Machine Elements I**, Prof. A. Daneshmehr *Spring 2025*
- **Robotics and Mechatronics**, Prof. M. Tale Masouleh *Fall 2024*
- **Design of Machine Elements II**, Prof. A. Daneshmehr *Fall 2024*
- **Numerical Calculations**, Prof. A. Fahim Spring 2024
- **Mechanics of Materials I**, Prof. M. Mahnama *Fall 2023*
- **Physics II**, Prof. S. Mohammadi *Spring and Fall 2023*

Selected Projects

Kinematic Modeling of Bio-Inspired Hyper-Redundant Gough-Stewart Robots *2025*
B.Sc. Thesis | Keras, PyTorch, Genetic Algorithm
- Trained neural networks for forward kinematics of multiple serially stacked GS platforms
- Designed a constrained redundancy-resolution algorithm for trajectory tracking and power consumption minimization.

Real-Time Motion Capture and IMU Game Controller *2024*
MediaPipe, OpenCV, Arduino
- Calibrated roll, pitch, and yaw measurements and mapped MPU-6050 tilt to a PyGame controller.
- Implemented smoothed, real-time leg-gesture recognition with MediaPipe.

Object Detection, Segmentation, and Robotic Grasp Point Generation *2024*
YOLOv5, FastSAM, Roboflow, OpenCV
- Trained detection and segmentation models and generated grasp poses from minimum-area rotated rectangles.

Robot-Type Image Classification with Convolutional Neural Networks *2024*
Python, PyTorch, ScikitLearn, CV2
- Collected and preprocessed an image-dataset and trained a CNN to classify five robot categories.

Capacitive Proximity Sensor for Liquid Level Measurement *2023*
Arduino, Proteus, Fritzing
- Designed, fabricated, optimized, and calibrated a high-sensitivity capacitive proximity sensor.

Selected Courses

- **Robotics and Mechatronics** 19.8/20
Prof. Mehdi Tale Masouleh – 2024
- **Artificial Intelligence** 17.0/20
Prof. Hakimeh Fadaei – 2023
- **Automatic Control** 17.6/20
Prof. Ali Sadighi – 2023
- **Medical Robotics*** 17.5/20
Prof. Saeed Behzadipour – 2026
- **Spine Biomechanics*** 18.3/20
Prof. Navid Arjmand – 2026
- **Optimal Design*** 18.0/20
Prof. Saeed Khodaygan – 2026

Honors and Awards

- Ranked 120th among 155,000+ participants in Mathematics and 60th among 174,000+ participants in Foreign Languages in Iran's Nationwide University Entrance Exam *2020*
- Ranked 180th among 2000+ participants in Nationwide Entrance Exam for a Master's Degree in Mechanical Engineering *2025*

Technical Skills

Programming: Python, MATLAB, C++, ROS, Arduino, HTML/CSS, JavaScript, Markdown, LaTeX
Technologies/Frameworks: Git, Linux, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Django
Engineering: SolidWorks, AutoCAD, ANSYS, Abaqus, Simulink, Adams, Fritzing, Proteus

Languages and Certifications

English: IELTS 8.0 (Reading: 9.0, Listening: 9.0, Speaking: 6.5, Writing: 6.5) **Persian:** Native
Coursera: Supervised Machine Learning: Regression and Classification (2022)
TopLearn: Programming with Python (2022)

References

Dr. Saeed Behzadipour, Professor at Sharif University of Technology behzadipour@sharif.edu
Dr. Mehdi Tale Masouleh, Associate Professor at University of Tehran m.t.masouleh@ut.ac.ir
Dr. Mohammad Ali Nazari, Assistant Professor at University of Tehran manazari@ut.ac.ir

*Graduate-level course